Engineering Economics (CSE-481) week-5 : Engineering Economics Concepts

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The Project Life Cycle — from Idea to Reality

- Every engineering project, whether it's a new data center or a mobile app, follows a **life cycle**.
- It begins with a **concept**, moves through **Design**, **Execution**, **Commissioning (Deployment),** and finally enters **Operation and Maintenance**.

The Project Life Cycle — from Idea to Reality

In the **concept stage**, we're full of ideas — it's the easiest and cheapest time to make changes.

But as the project moves forward — into **design**, **execution**, and **operation** — **changes** become harder and much more expensive.

That's why engineers say: **The earlier you think, the cheaper you learn.**

Feasibility Study — Asking 'Is It Worth It?'

Before writing a single line of code or buying a single piece of hardware, we perform a **feasibility study**.

It's a deep investigation that asks: *Is this project really worth doing?*

We examine three major dimensions:

- **Market Feasibility:** Is there demand for this product or solution?
- **Technical Feasibility:** Do we have the technology and skills to build it?
- **Economic Feasibility:** Will the benefits outweigh the costs?

A good feasibility study helps us:

- Narrow down business alternatives,
- Discover new opportunities,
- Identify reasons not to proceed,

In short, **it saves both time and money**, by helping us choose the best path forward.

What Is Engineering Economic Analysis?

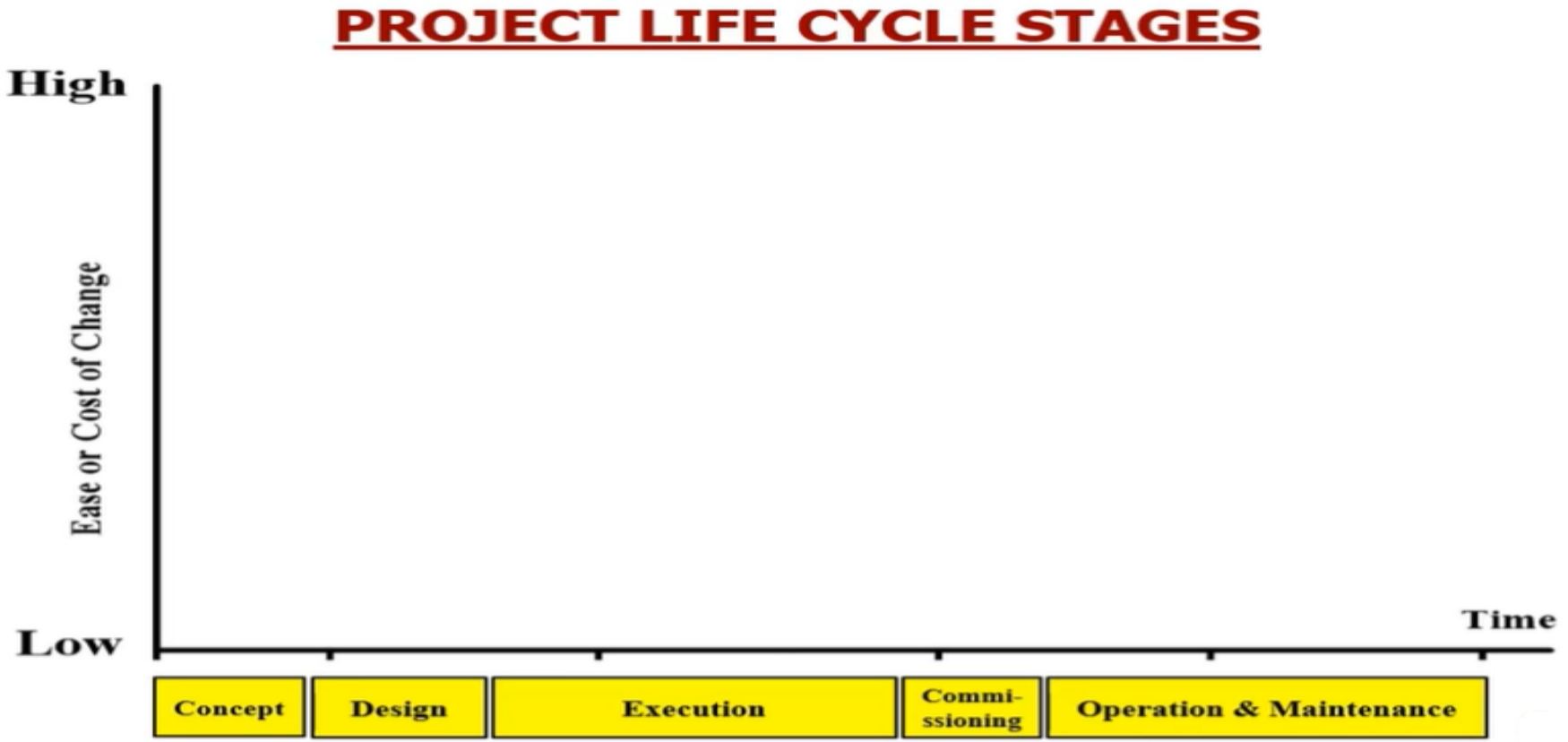
Every design we create requires **money** to design, build, and operate.

Engineering Economic Analysis means evaluating all **costs, revenues, and benefits** over the lifetime of a project to determine if it's worthwhile.

It helps us answer key questions such as:

- Which projects should be prioritized?
- Where should we locate the system?
- How should it be designed, operated, or maintained?
- Should we accept or reject the project altogether?

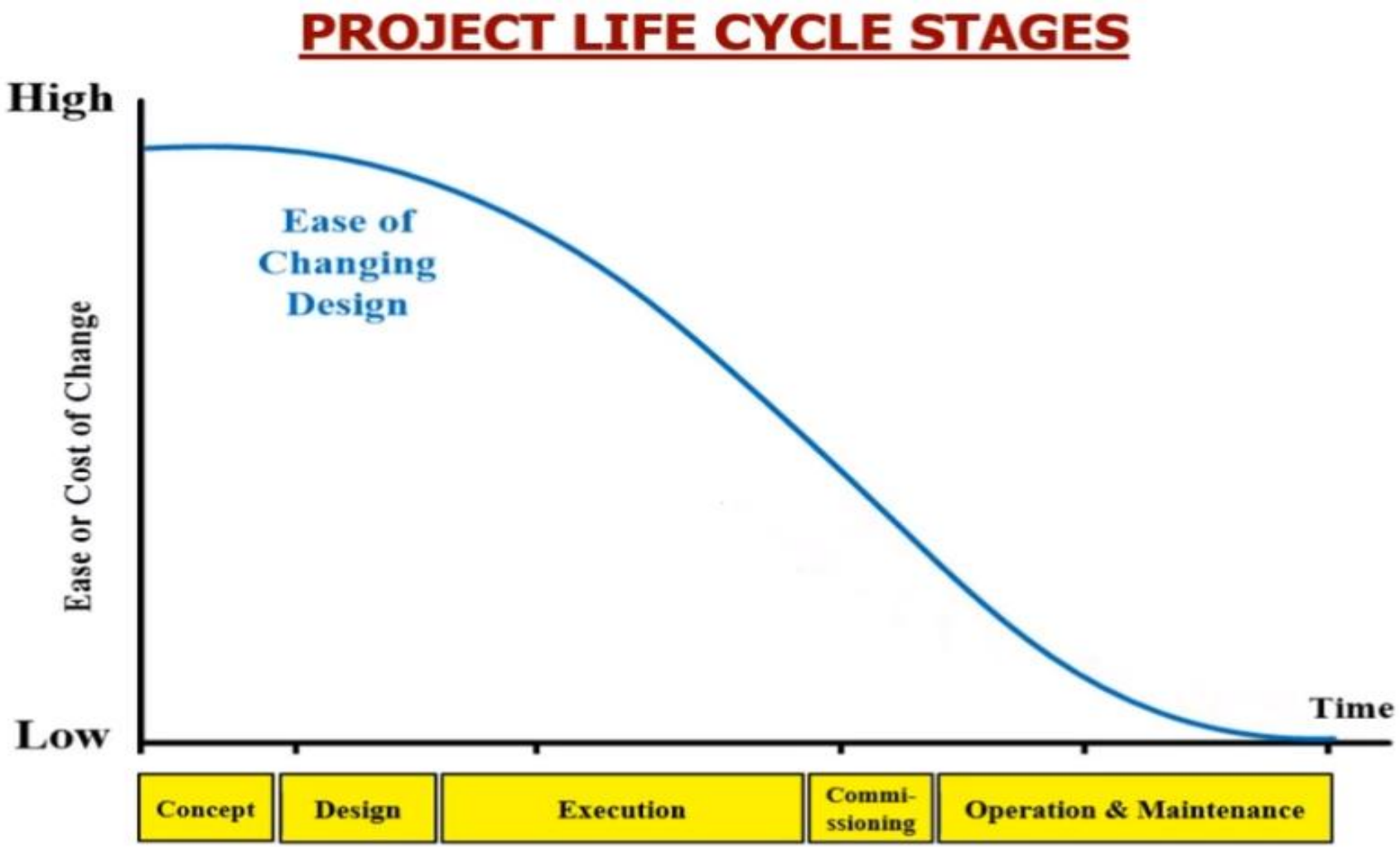
Project Life Cycle Stages



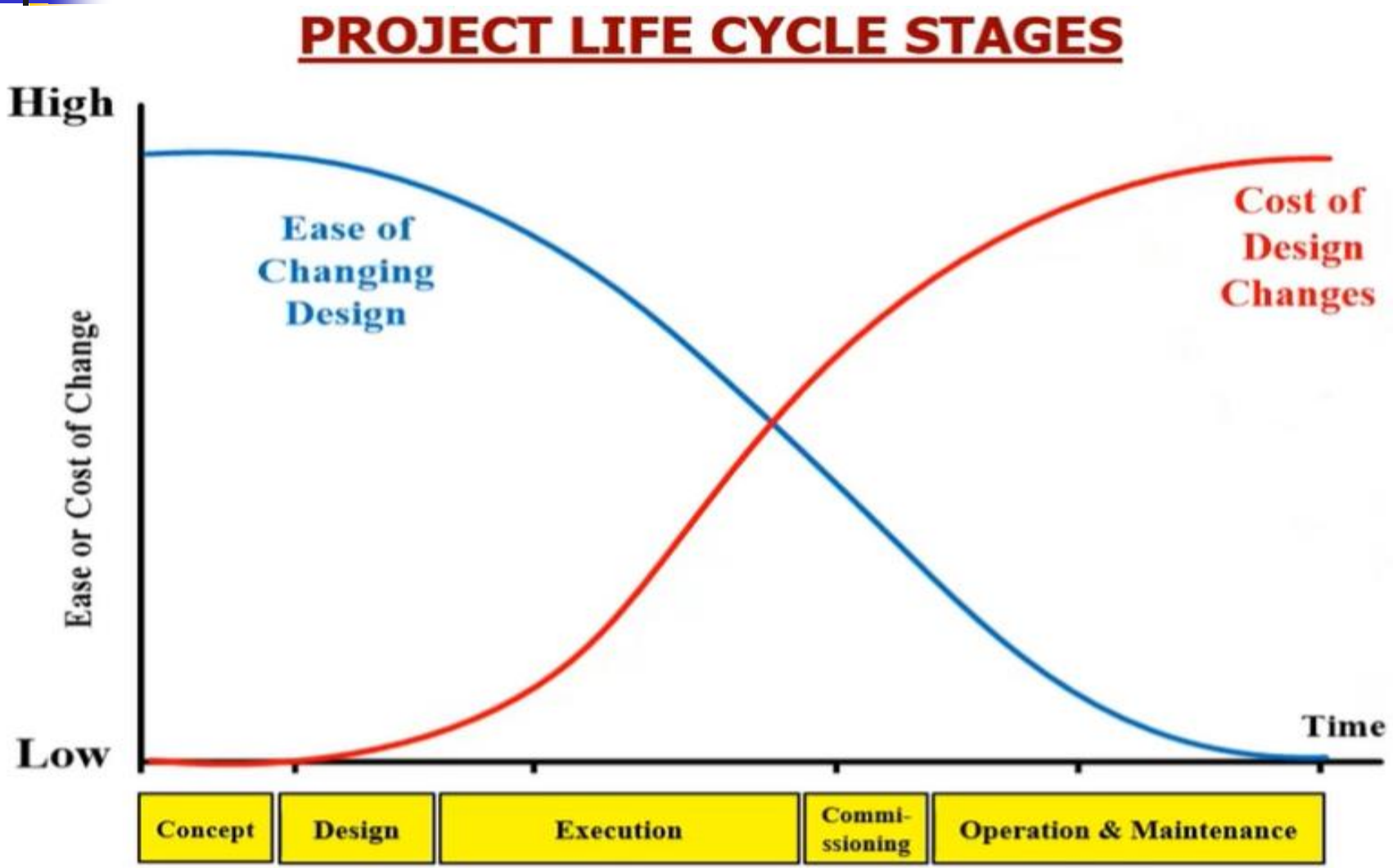
The economic investment decision

- 1. Q : These will be an adequate return out of this investment or not ?
- 2. Q : Does it make sense to invest Money in the project.

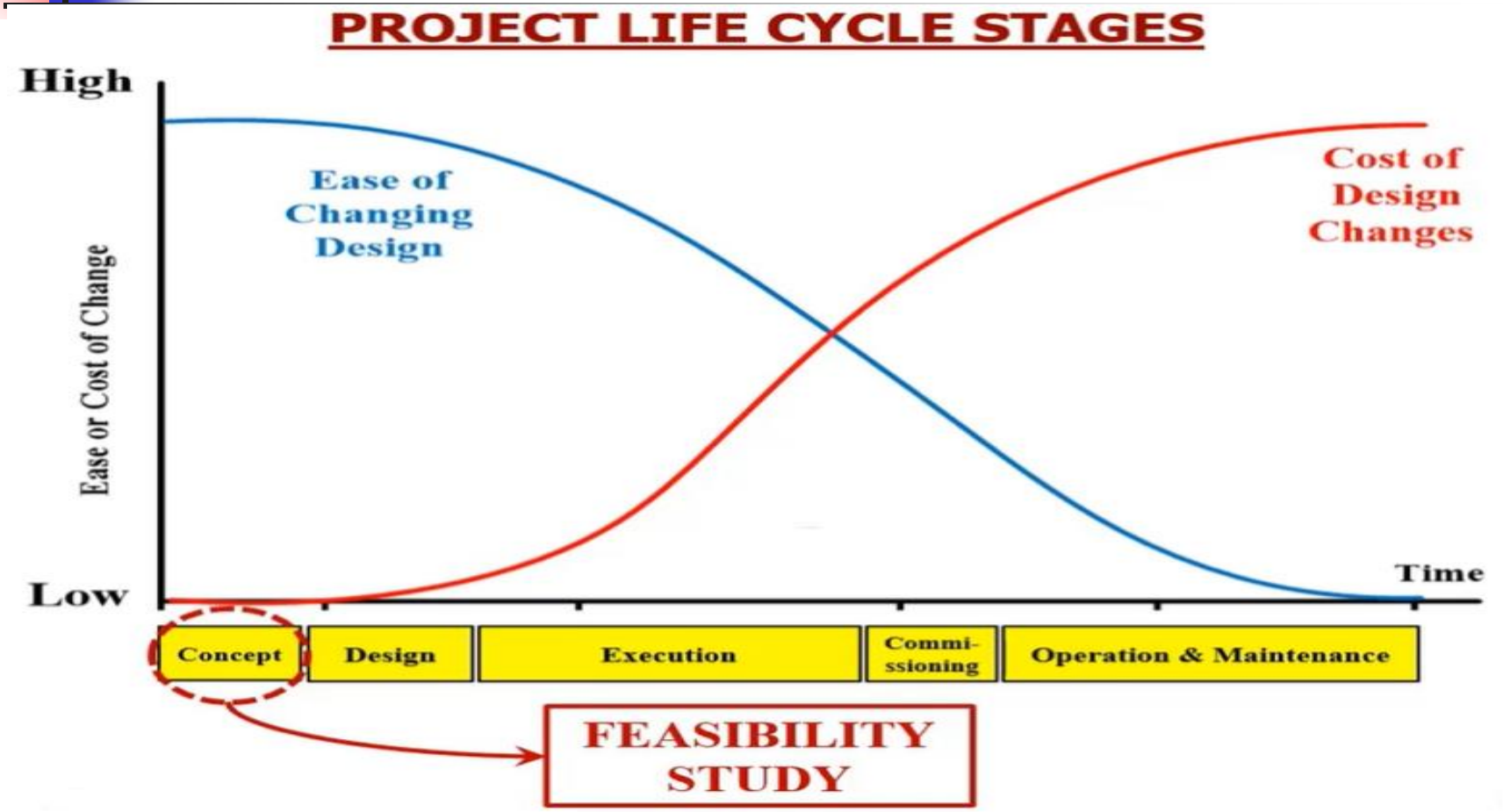
Project Life Cycle Stages



Project Life Cycle Stages



Project Life Cycle Stages



the concept stage is the most important factor that affects the success of every other stages and the other stages sucess will depends on the our first stage conceptual stage. In the concept phase, we usually do a feasibility study.

Feasibility Study Process

> " A typical feasibility study is an analysis aimed at discovering whether or not a specific project will be successful and thus provides an adequate return to the investor "
>
> *William C. Frank, " You Can Increase the Worth of Feasibility Studies " Infosystems, Sept. 1981*

In this definition it is very obvious that we are talking about the successfulness of the project and to provide an adequate return.

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"Feasibility analysis is simply an application of a basic management concept: evaluate the impact of a decision before you make it "

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" Feasibility analysis is the final pre-investment study phase of a project's life cycle and should contain all the information necessary to help decision-makers make rational decisions"

March, W.M., " Environmental Analysis for Land Use & Site Planning " McGraw-Hill, 1978

All these definition says that feasibility study is concerned with whether it is worth to invest in the project or not.

Why to Do a Feasibility Study ?

- Narrow down business alternatives
- Introduce **new opportunities** through the investigation process
- **Identifies reasons not to proceed**. For example we are not proceeding with the project or we don't think this project is profitable (doing the feasibility analysis is a very good reason why or why not to proceed with such a project)
- Provides documentation that the business project was thoroughly investigated, thus:
  - Helps to increase investment in the company
  - Helps in **securing funding from lending institutions** and other sources

Feasibility Study Process

identify if we have the capability to **conduct this project** or **not or invest** in this project or not.

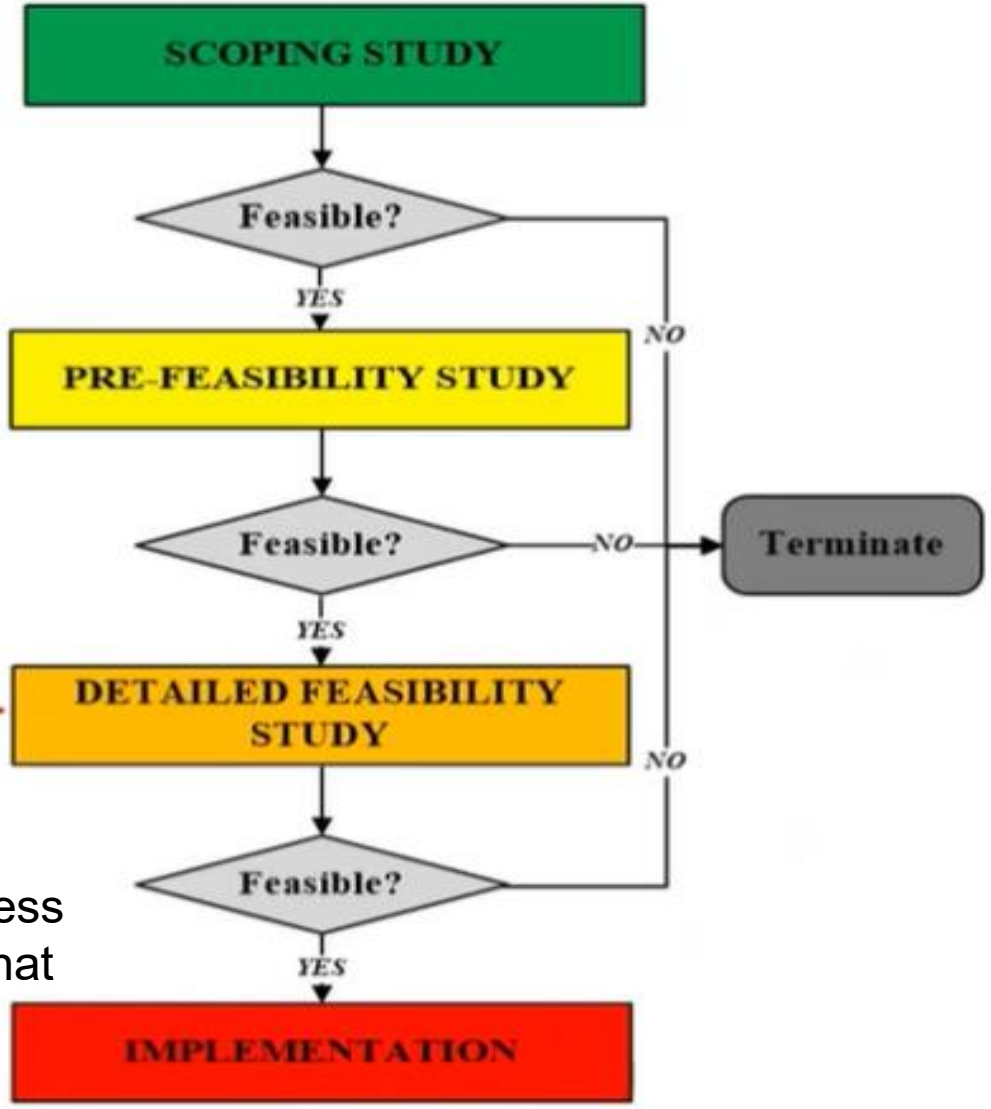
determine, analyze and select the best identified alternatives in order to save time and save money

Market Feasibility

Technical Feasibility

Economical Feasibility

the measurement of the cost effectiveness of the project (What are benefits and what are the costs of the project)



What is Engineering Economic Analysis ?

- Every engineer design will **require the money** and **engineer design need to spend money** whether in the **design phase** and whether in the **execution phase**. After the execution and implementation stages we are expecting **revenues or benefits**, as well as **operating costs** that will usually occur for years throughout the life cycle.

- Engineering economic analysis indicates the analysis of the **costs, revenues and benefits**.

What is Engineering Economic Analysis ?

- Thus, the economic analysis of costs, revenues, and benefits occurring over time for engineering projects is called **engineering economic analysis**

What is Engineering Economic Analysis ?

- Engineering economic analysis is used to answer many project and project-related questions such as:

  - *Which engineering projects are worthwhile and which should have a higher priority?*
  - *Where to locate the project ?*
  - *How should the engineering project be designed ?*
  - *How should the engineering project be operated and maintained ?*
  - *Accept or reject the project ?*
  - *Lease or buy ?*

What is Engineering Economic Analysis ?

- Engineering economic analysis can also be used to answer personal questions such as:

  - *How to achieve a long-term financial goals?*
  - *How to compare different ways to finance purchases ?*
  - *How to make short and long term investment decisions ?*

Cost Terminologies

- **Fixed Costs:**
- Costs that are constant regardless of the level of output or activity (e.g. Administrative expenses, rental, insurance, etc.)

- **Variable Costs:**
- Costs that change depending on the level of output or activity (e.g. material, labor, energy consumption, etc.)

Cost Terminologies

- **Recurring Costs:**
- Refer to any expense that is known and anticipated and that occurs at regular intervals (e.g. annual regular operation and maintenance cost, annual rentals etc.)

- **Non-Recurring Costs:**
- The expenses that occur at irregular intervals and are sometimes difficult to plan for or predict from a budgeting perspective (e.g. emergency maintenance cost, replacement cost, etc.)

Cost Terminologies

- **Sunk Costs:**

- Money that is already spent as a result of a past decision

- Sunk costs should be disregarded in engineering economic analysis because current decisions cannot change the past.

- As engineering economists we deal with present and future opportunities

Cost Terminologies

- **Opportunity Costs:**

- The benefit that is **forgone** by engaging a business resource in a chosen activity instead of engaging that same resource in a **forgone** activity

- In simplified terms, it is **the cost of what else one could have chosen to do**

- **the cost of alternative is considered** as **your opportunity cost**.

- In the simple case, you have 2 alternatives and you have chosen alternative one so the cost of alternative is considered as your opportunity cost.

The Time Value of Money — Why 'Now' Beats 'Later'

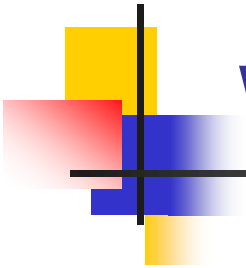
Would you rather receive $1,000 today, or $1,000 one year from now?

Most people choose today — and they're right.
Because **money today** can be invested, can grow, and has **earning power over time**.

This is the **Time Value of Money (TVM)** — the principle that drives the entire economic analysis model.

- **Present Value (P):** What future money is worth today
- **Future Value (F):** What today's money will be worth in the future

Even in software projects, this matters.
For example:
- Investing in faster servers now might reduce downtime costs later.
- Spending on cloud optimization could save thousands in monthly fees.



Why It Matters for Computer Engineers

In computer science, economics is not just theory — it's strategy.

When you design a system, you're not only optimizing *performance*, you're optimizing *cost-efficiency* over time.

- Should your startup build its own AI infrastructure or use AWS?
- Should your company invest in new servers or migrate to the cloud?
- Should you refactor legacy code or start fresh?

It's about **sustainability, strategy, and smart use of resources**.

Time Value of Money

- The idea that the money received today is worth more than the money received in the future.

- **Time Value of Money** is the principle that drives the expected return model.

Applying Time Value of Money

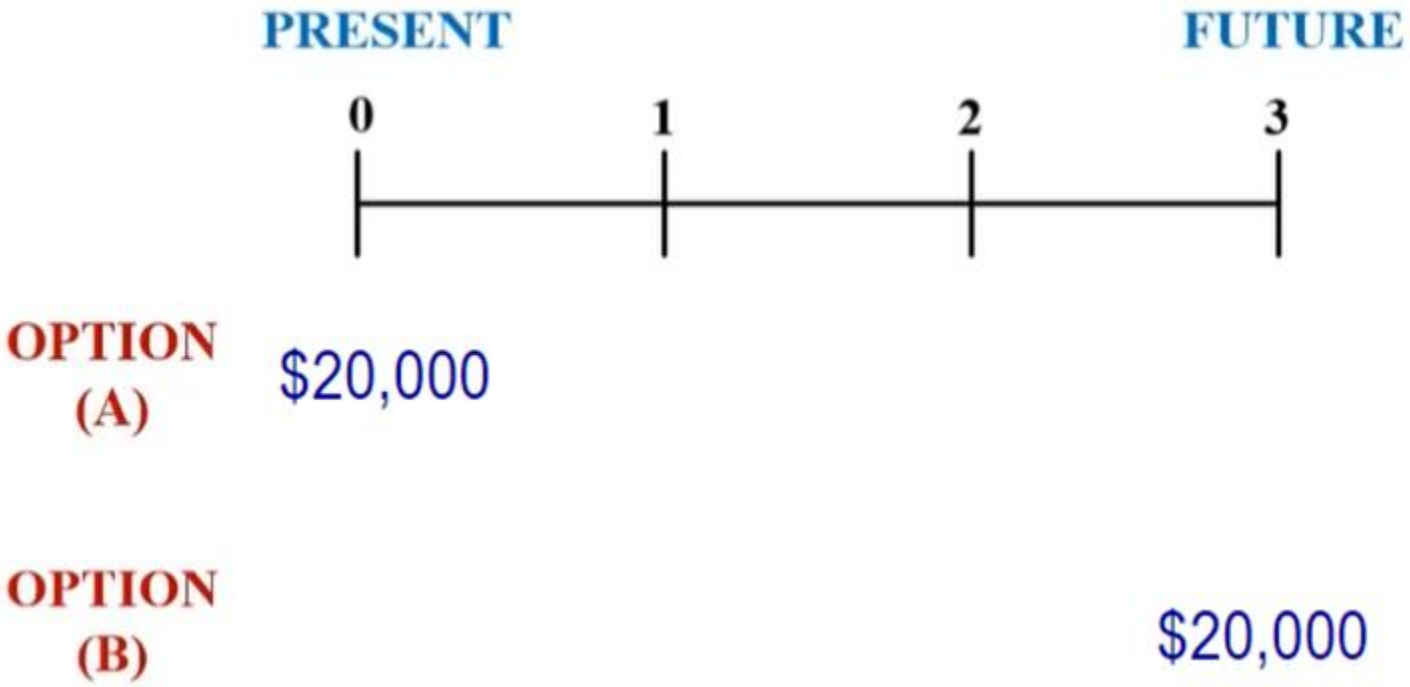
1) *Congratulations!* You have won a cash prize of $20,000! You have two options. Which is better?

A. Receive $20,000 now

B. Receive $20,000 in 3 years

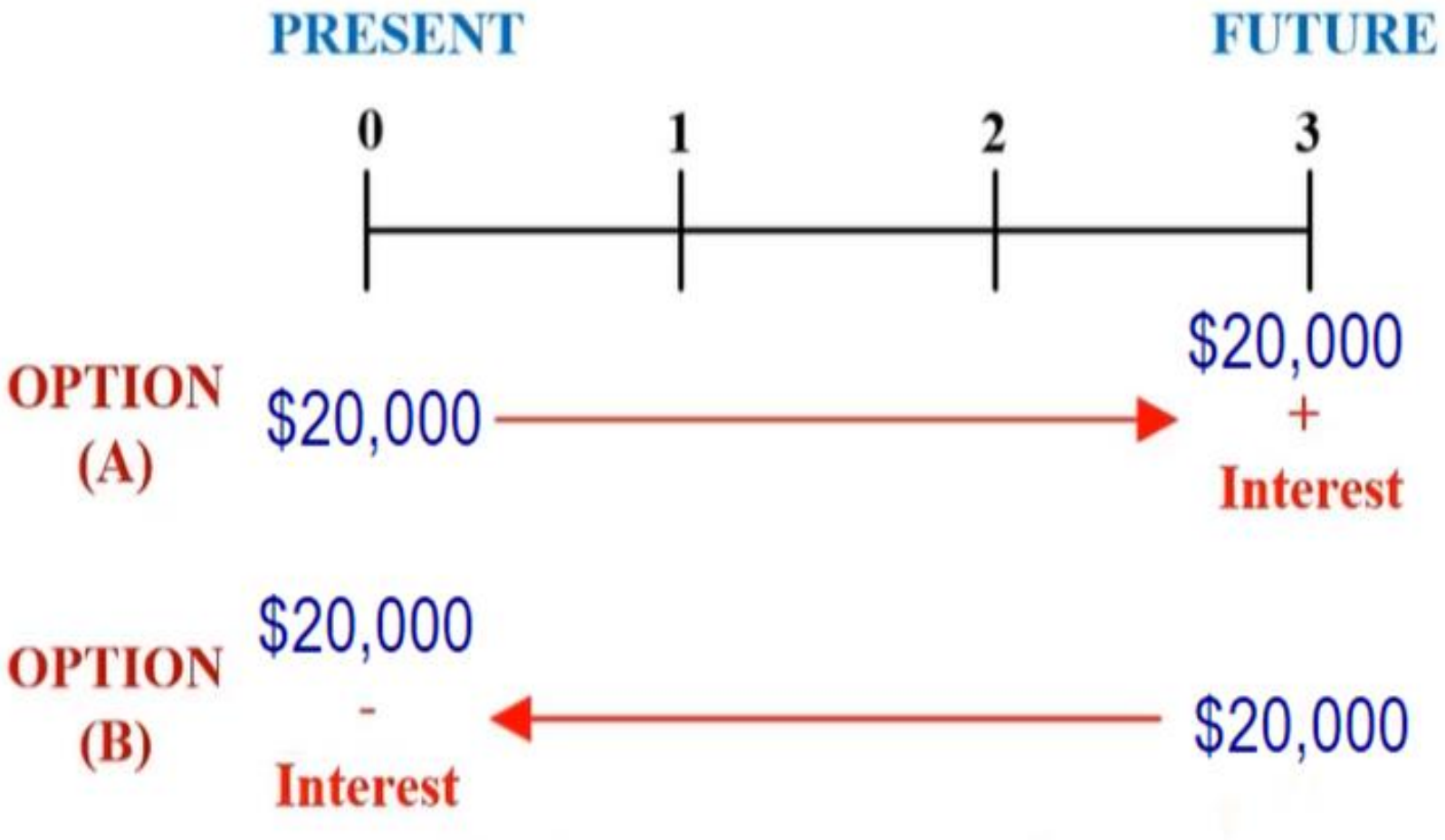
The choice is clear: most people will choose to take the money today.

Time Value of Money



Earning money today is better than earning money later.

Time Value of Money



Applying Time Value of Money

1) ***Congratulations!*** You have won a cash prize of $20,000! You have two options. Which is better?

A. Receive $20,000 now

B. Receive $20,000 in 3 years

The choice is clear: most people will choose to take the money today.

2) Now, what if the options are:

A. Receive $20,000 now

Or

B. Receive $23,000 in 3 years

Here, the choice is less clear. The option you would choose is based on how much interest you think you can earn each year on $20,000 received today

Time Value of Money

- The time value of money is the concept that money available at the present time is worth more than the same amount in the future due to its potential earning capacity.

Money has an earning power with time



Time Value of Money

You have 3 options

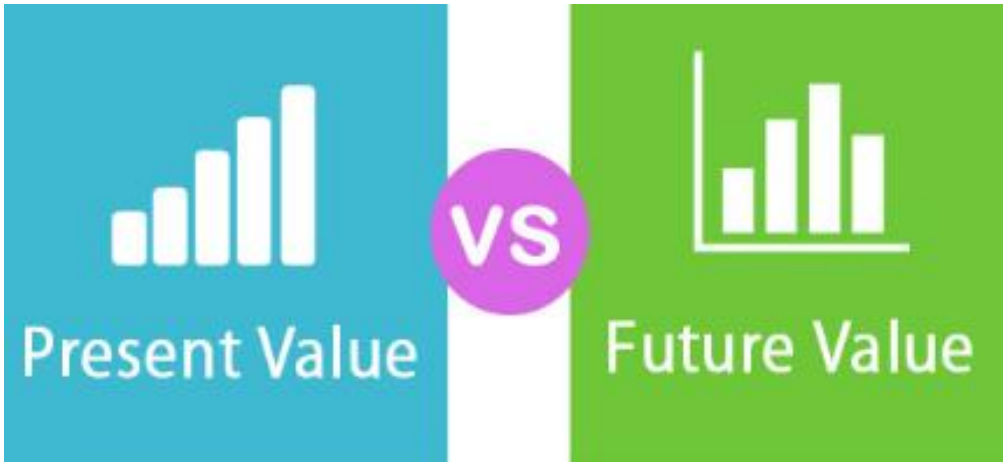
Would you prefer to:

- (A) Receive $10.000 now ?
- (B) Receive $ 4.000 at the end of year 1, 2, and 3 ?
- (C) Receive $15.000 three years from now ?

Time Value of Money

Present Value (P): The current value of a future amount of money, or a series of payments, evaluated at a given interest rate

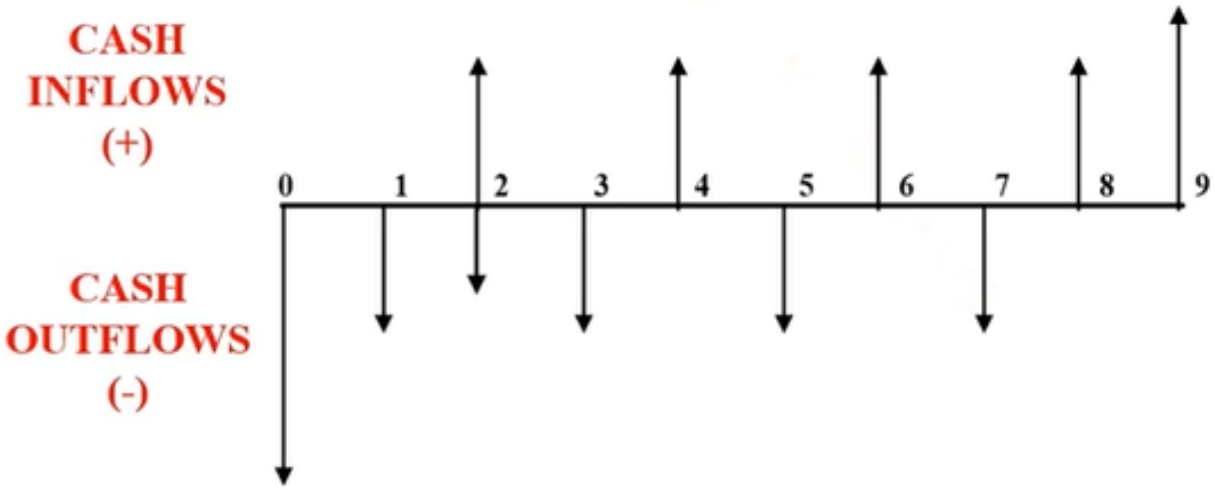
Future Value (F): The value at a certain future time of a present amount of money, or a series of payments, evaluated at a given interest rate



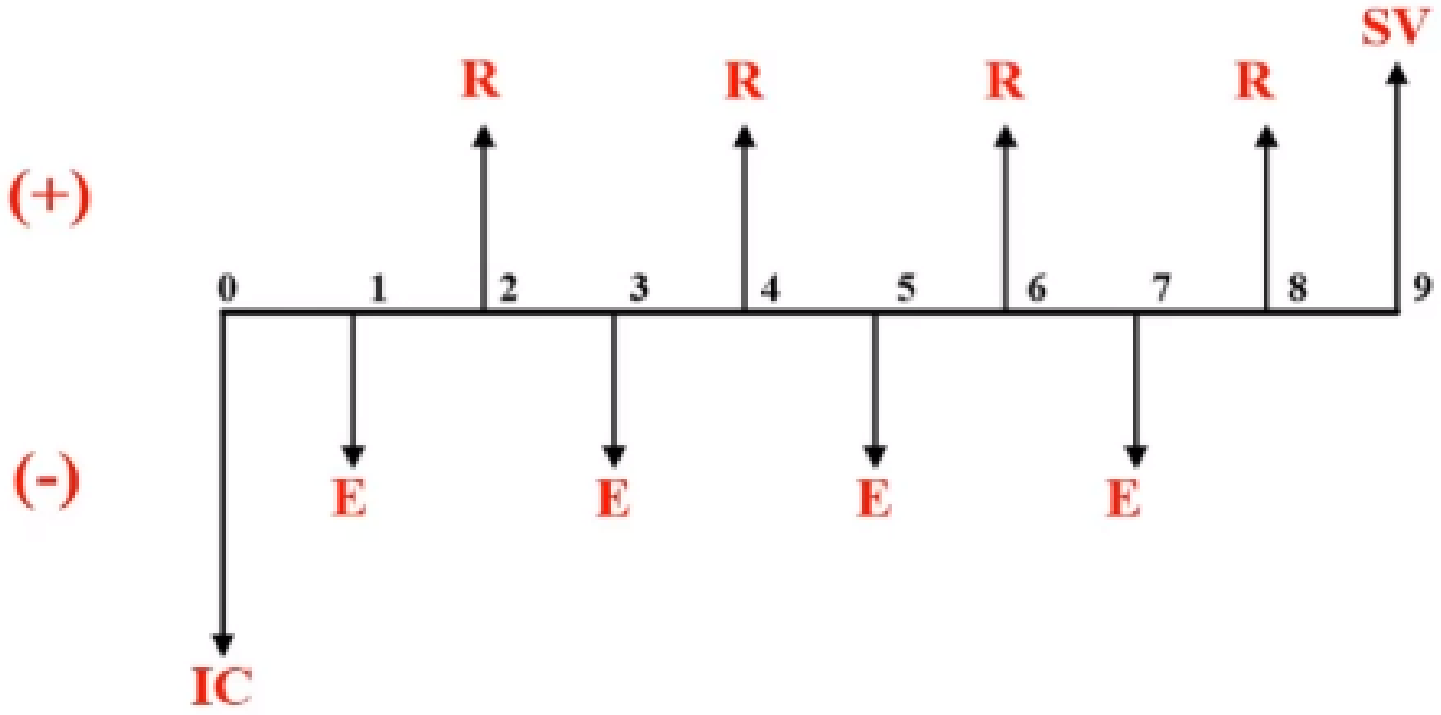
Cash Flow Diagram

- The costs and benefits of engineering projects occur over time and are represented on cash flow diagrams

- In engineering economic analysis all costs and benefits are assumed to occur at the end of each period



Cash Flow Diagram



- **IC** (Initial Cost or First Cost): expense to build or buy and install
- **E** (Expenses): operation, maintenance, labor costs, etc.
- **R** (Revenues): receipts due to sale of products or services
- **SV** (Salvage Value): resale value of an asset at the end of its useful life

Interest and Interest Rate

- The interest is a measure of the increase between the original sum of money (borrowed or invested) and the final sum of money (owed or accrued)
- The original amount is referred to as ***Principal***
- Thus, the interest rate reflects the fact that money has a ***time value***

Interest Amount = Final Amount – Original Amount

Interest and Interest Rate

If you deposit $1,000 today in a bank at an interest rate of 10% per year. How much will you have after one year?

Interest Amount = 1000 x 0.10 = $100

Final Amount (after one year) = 1000 + 100 = **$1,100**

Types of Interest Rate

- **SIMPLE INTEREST:**
  - Interest is charged on only the original amount (principal amount)
- **COMPOUND INTEREST:**
  - Each period (e.g. year), interest is charged based on the total amount owed or accrued at the end of the previous period



Simple Interest

Period	Amount at the Beginning of Period	Interest for Period	Amount at the End of Period
1			
2			
3			
4			
-			
n			

Simple Interest

Period	Amount at the Beginning of Period	Interest for Period	Amount at the End of Period
1	P	Pi	P(1 + i)
2	P(1 + i)	Pi	P(1 + 2i)
3	P(1 + 2i)	Pi	P(1 + 3i)
4	P(1 + 3i)	Pi	P(1 + 4i)
-	-	-	-
n	P(1 + (n-1)i)	Pi	P(1 + ni)

$$F_n = P(1 + ni)$$ → For "Simple" Interest

Simple Interest

Assume that you deposit $1,000 in an account earning 5% simple interest rate per year for three years. How much will you have after three years?

Period	Amount at the Beginning of Period	Interest for Period	Amount at the End of Period
1	1000	50	1050
2	1050	50	1100
3	1100	50	1150

Compound Interest

Period	Amount at the Beginning of Period	Interest for Period	Amount at the End of Period
1	P	Pi	$P(1 + i)$
2	$P(1 + i)$	$P(1 + i)i$	$P(1 + i)^2$
3	$P(1 + i)^2$	$P(1 + i)^2 i$	$P(1 + i)^3$
4	$P(1 + i)^3$	$P(1 + i)^3 i$	$P(1 + i)^4$
-	-	-	-
n	$P(1 + i)^{n-1}$	$P(1 + i)^{n-1} i$	$P(1 + i)^n$

$$F_n = P(1 + i)^n$$ → For "Compound" Interest

Compound Interest

Assume that you deposit $1,000 in an account earning 5% compound interest rate per year for three years. How much will you have after three years?

$$F_n = P(1 + i)^n$$

$$F_3 = 1000(1 + 0.05)^3 = \$1{,}157.625$$

Period	Amount at the Beginning of Period	Interest for Period	Amount at the End of Period
1	1000	50	1050
2	1050	52.5	1102.5
3	1102.5	55.125	1157.625

Compound Interest

Assume that you deposit $1,000 in an account earning 7% interest rate per year for thirty years. How much will you have after thirty years?

Simple Interest

$F_{30} = 1000 \times (1 + (0.07 \times 30)) =$ **$3,100**

Compound Interest

$F_{30} = 1000 \times (1 + 0.07)^{30} =$ **$7,612.3**

Compound Interest

